Exploratory Data Analysis — SMAP (Soil Moisture Active Passive)

Dataset SMAP berasal dari data telemetri satelit SMAP milik NASA. Memiliki format yang sama dengan MSL namun mencakup 55 channel dengan karakteristik sinyal yang berbeda.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import matplotlib.gridspec as gridspec
from pathlib import Path
import ast, warnings, os
warnings.filterwarnings('ignore')

plt.rcParams.update({
    'figure.dpi': 120,
    'font.size': 10,
    'axes.titlesize': 11,
    'axes.labelsize': 10,
    'lines.linewidth': 0.9,
    'axes.grid': True,
    'grid.alpha': 0.3,
})

BASE = Path(r'D:\CARLA')
COLORS = {'normal': '#2196F3', 'anomaly': '#FF5722', 'val': '#4CAF50'}

# Buat folder output jika belum ada
FIG_DIR = BASE / 'results' / 'figures'
FIG_DIR.mkdir(parents=True, exist_ok=True)
print(f'Setup selesai. Output gambar: {FIG_DIR}')
```

```
Setup selesai. Output gambar: D:\CARLA\results\figures
```

1. Memuat Metadata

```
DATA_DIR = BASE / 'datasets' / 'MSL_SMAP'
df_labels = pd.read_csv(DATA_DIR / 'labeled_anomalies.csv')
smap = df_labels[df_labels['spacecraft'] == 'SMAP'].copy().reset_index(drop=

print(f"Jumlah channel SMAP : {len(smap)}")
display(smap[['chan_id','class','num_values','anomaly_sequences']].head(10))
```

```
Jumlah channel SMAP : 55
```

	chan_id	class	num_values	anomaly_sequences
0	P-1	[contextual, contextual, contextual]	8505	[[2149, 2349], [4536, 4844], [3539, 3779]]
1	S-1	[point]	7331	[[5300, 5747]]
2	E-1	[contextual, contextual]	8516	[[5000, 5030], [5610, 6086]]
3	E-2	[point]	8532	[[5598, 6995]]
4	E-3	[point]	8307	[[5094, 8306]]
5	E-4	[point]	8354	[[5450, 8261]]
6	E-5	[point]	8294	[[5600, 5920]]
7	E-6	[point]	8300	[[5610, 5675]]
8	E-7	[point]	8310	[[5394, 5674]]
9	E-8	[point]	8532	[[5400, 6022]]

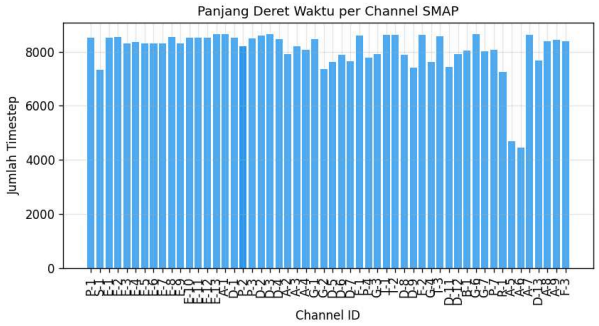
2. Distribusi Panjang Deret Waktu

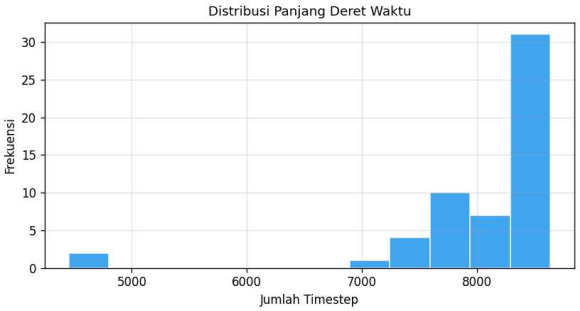
```
lengths = smap['num_values'].values
fig, axes = plt.subplots(1, 2, figsize=(14, 4))

axes[0].bar(smap['chan_id'], lengths, color=COLORS['normal'], alpha=0.8)
axes[0].set_xlabel('Channel ID')
axes[0].set_ylabel('Jumlah Timestep')
axes[0].set_title('Panjang Deret Waktu per Channel SMAP')
axes[0].tick_params(axis='x', rotation=90)

axes[1].hist(lengths, bins=12, color=COLORS['normal'], edgecolor='white', al
axes[1].set_xlabel('Jumlah Timestep')
axes[1].set_ylabel('Frekuensi')
axes[1].set_title('Distribusi Panjang Deret Waktu')

plt.tight_layout()
plt.savefig(BASE / 'results' / 'figures' / 'eda_smap_lengths.png', bbox_inch
plt.show()
print(pd.Series(lengths).describe().round(1).to_string())
```






```
count      55.0
mean     8073.4
std       795.5
min      4453.0
25%      7895.5
50%      8307.0
75%      8515.0
max      8640.0
```

3. Distribusi Tipe Anomali

```
import re
import ast
from collections import Counter

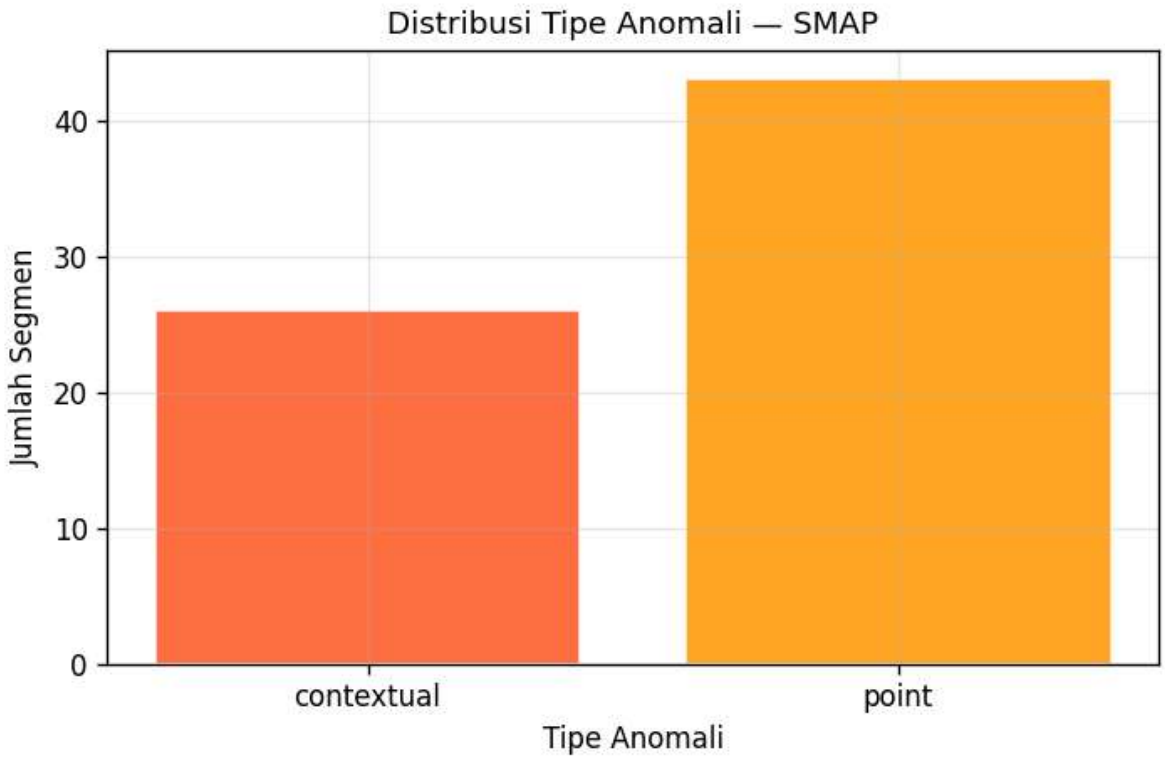
def parse_class(c):
    return re.findall(r'[A-Za-z_][A-Za-z0-9_]*', str(c))

def parse_seqs(s):
    return ast.literal_eval(s)

all_types = []
for _, row in smap.iterrows():
    all_types.extend(parse_class(row['class']))

type_counts = Counter(all_types)

fig, ax = plt.subplots(figsize=(6, 4))
ax.bar(type_counts.keys(), type_counts.values(),
       color=[COLORS['anomaly'], '#FF9800'], edgecolor='white', alpha=0.85)
ax.set_xlabel('Tipe Anomali')
ax.set_ylabel('Jumlah Segmen')
ax.set_title('Distribusi Tipe Anomali — SMAP')
plt.tight_layout()
plt.savefig(BASE / 'results' / 'figures' / 'eda_smap_anomaly_types.png', bbo
plt.show()
for k, v in type_counts.items():
    print(f"  {k:15s}: {v} segmen")
```




```
contextual     : 26 segmen
point          : 43 segmen
```

4. Visualisasi Sampel Channel

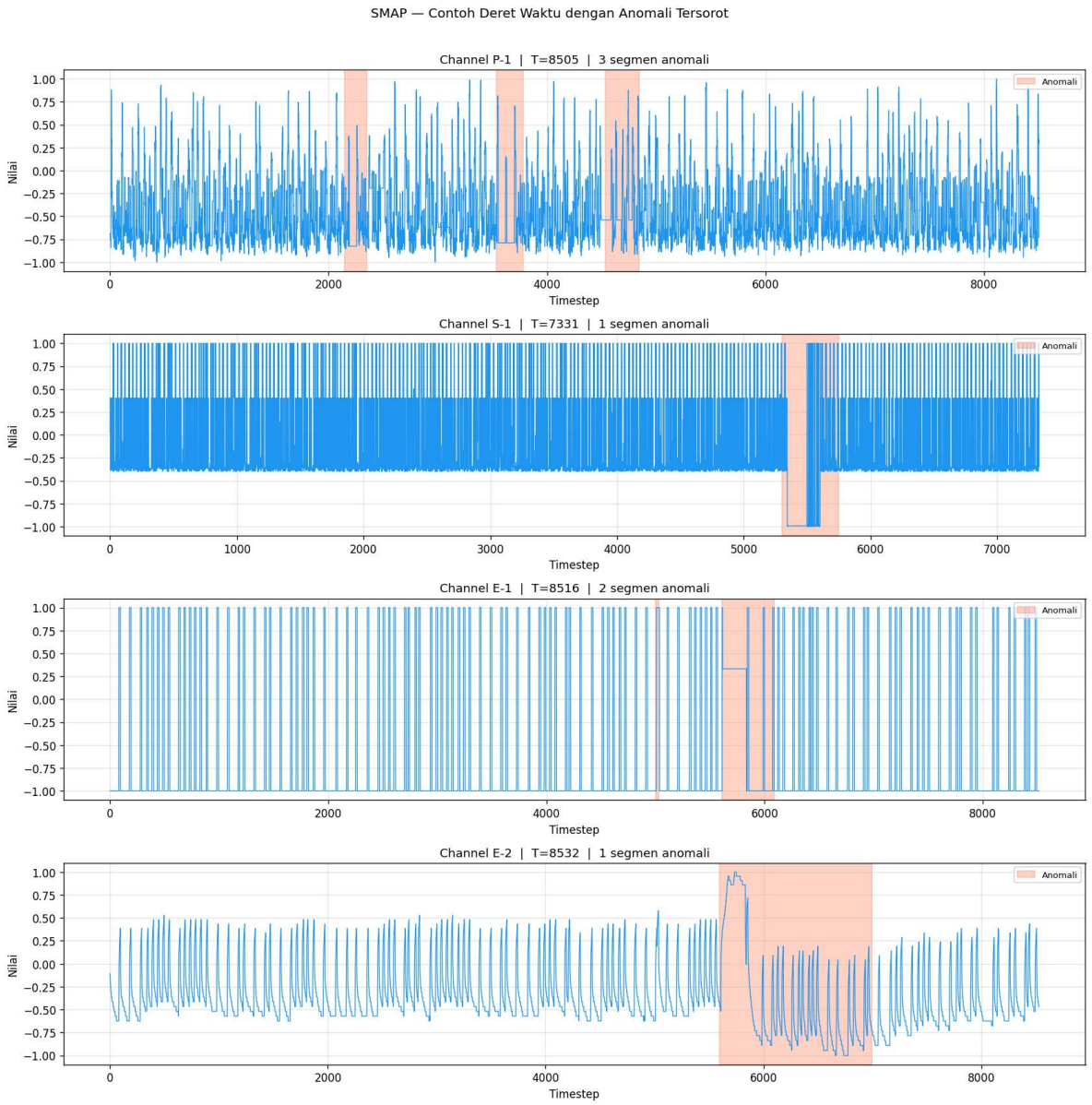
```
def load_smap(chan_id, split='test'):
    return np.load(DATA_DIR / 'data' / split / f'{chan_id}.npy')

sample_chans = smap['chan_id'].iloc[:4].tolist()
fig, axes = plt.subplots(len(sample_chans), 1, figsize=(14, 3.5 * len(sample

for ax, chan_id in zip(axes, sample_chans):
    row = smap[smap['chan_id'] == chan_id].iloc[0]
    data = load_smap(chan_id, 'test')
    T = data.shape[0]
    ax.plot(np.arange(T), data[:, 0], color=COLORS['normal'], linewidth=0.8)
    seqs = parse_seqs(row['anomaly_sequences'])
    for i, (s, e) in enumerate(seqs):
        ax.axvspan(s, e, color=COLORS['anomaly'], alpha=0.25,
                   label='Anomali' if i == 0 else '_')
    ax.set_title(f'Channel {chan_id}  |  T={T}  |  {len(seqs)} segmen anomal
    ax.set_xlabel('Timestep')
    ax.set_ylabel('Nilai')
    ax.legend(fontsize=8, loc='upper right')

plt.suptitle('SMAP — Contoh Deret Waktu dengan Anomali Tersorot', fontsize=1
plt.tight_layout()
```

```
plt.savefig(BASE / 'results' / 'figures' / 'eda_smap_timeseries.png', bbox_i
plt.show()
```

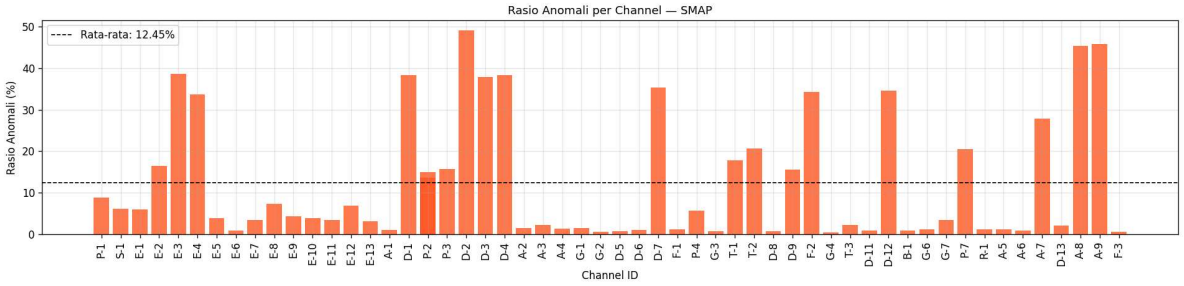


5. Rasio Anomali per Channel

```
anomaly_ratios = []
for _, row in smap.iterrows():
    T = row['num_values']
```

```
    seqs = parse_seqs(row['anomaly_sequences'])
    n_anom = sum(e - s + 1 for s, e in seqs)
    anomaly_ratios.append(n_anom / T * 100)
smap['anomaly_ratio_%'] = anomaly_ratios

fig, ax = plt.subplots(figsize=(16, 4))
ax.bar(smap['chan_id'], smap['anomaly_ratio_%'], color=COLORS['anomaly'], al
ax.axhline(np.mean(anomaly_ratios), color='black', linestyle='--', linewidth
           label=f'Rata-rata: {np.mean(anomaly_ratios):.2f}%')
ax.set_xlabel('Channel ID')
ax.set_ylabel('Rasio Anomali (%)')
ax.set_title('Rasio Anomali per Channel — SMAP')
ax.tick_params(axis='x', rotation=90)
ax.legend()
plt.tight_layout()
plt.savefig(BASE / 'results' / 'figures' / 'eda_smap_anomaly_ratio.png', bbo
plt.show()
print(f"Rata-rata rasio anomali : {np.mean(anomaly_ratios):.2f}%")
```




```
Rata-rata rasio anomali : 12.45%
```

6. Perbandingan Statistik Train vs Test (Channel P-1)

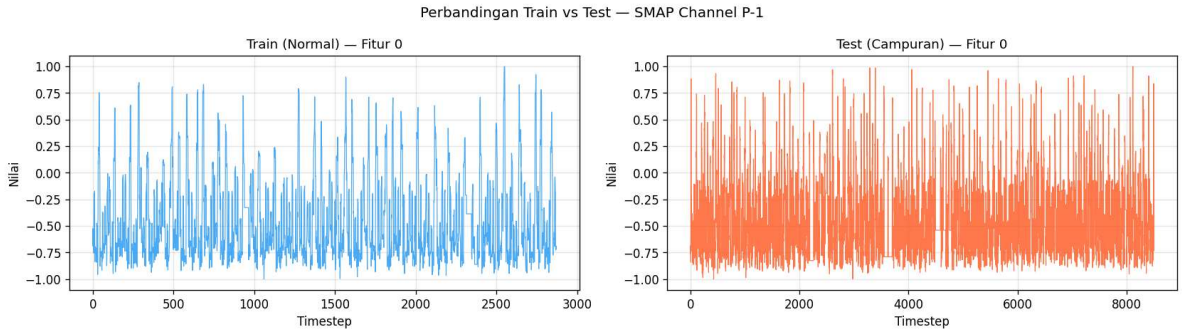
```
chan = 'P-1'
tr = load_smap(chan, 'train')
te = load_smap(chan, 'test')

print(f"Channel {chan}:")
print(f"  Train shape : {tr.shape}")
print(f"  Test shape  : {te.shape}")

fig, axes = plt.subplots(1, 2, figsize=(14, 4))
for i, (name, data, color) in enumerate([
        ('Train (Normal)', tr, COLORS['normal']),
        ('Test (Campuran)', te, COLORS['anomaly'])]):
    axes[i].plot(data[:, 0], color=color, linewidth=0.7, alpha=0.8)
    axes[i].set_title(f'{name} — Fitur 0')
    axes[i].set_xlabel('Timestep')
    axes[i].set_ylabel('Nilai')
plt.suptitle(f'Perbandingan Train vs Test — SMAP Channel {chan}')
plt.tight_layout()
```

```
plt.savefig(BASE / 'results' / 'figures' / 'eda_smap_train_test.png', bbox_i
plt.show()
```

```
Channel P-1:
  Train shape : (2872, 25)
  Test shape  : (8505, 25)
```



7. Ringkasan Dataset SMAP

```
print("=" * 50)
print("RINGKASAN DATASET SMAP")
print("=" * 50)
total_segs = sum(len(parse_seqs(r)) for r in smap['anomaly_sequences'])
print(f"Jumlah channel          : {len(smap)}")
print(f"Jumlah fitur per sample : {load_smap(smap.iloc[0]['chan_id'], 'test')
print(f"Rata-rata panjang test  : {smap['num_values'].mean():.0f} timestep")
print(f"Total segmen anomali    : {total_segs}")
print(f"Rata-rata rasio anomali : {np.mean(anomaly_ratios):.2f}%")
print(f"Tipe anomali            : point, contextual")
```

```
==================================================
RINGKASAN DATASET SMAP
==================================================
Jumlah channel          : 55
Jumlah fitur per sample : 25
Rata-rata panjang test  : 8073 timestep
Total segmen anomali    : 69
Rata-rata rasio anomali : 12.45%
Tipe anomali            : point, contextual
```